

Grade 6 Accelerated  
Unit 11  
Lesson 1 Homework

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Find the sum for each expression in questions 1-6.

1.  $8.45 + (-3.55)$

$4.9$  or  $\frac{49}{10}$  or  $4\frac{9}{10}$

4.  $-\frac{3}{8} + -\frac{21}{9}$

$-\frac{65}{24}$  or  $-2\frac{17}{24}$  or  $-2.708\bar{3}$

2.  $\left(-6\frac{2}{5}\right) + \left(\frac{7}{3}\right)$

$-4.0\bar{6}$  or  $-\frac{61}{15}$  or  $-4\frac{1}{15}$

5.  $(-91.125) + (2.98)$

$-88.145$  or  $-\frac{17629}{200}$  or  $-88\frac{29}{200}$

3.  $73.381 + 46.72$

$120.101$  or  $120\frac{101}{1000}$

6.  $14.31 + (-22.91)$

$-8.6$

Using the expression bank, find a matching expression for questions 7-10.

**Expression Bank**

$$\frac{8}{5} + \left(-7\frac{19}{20}\right) = \frac{-127}{20}$$

$$-\frac{13}{4} + 2\frac{5}{12} = \frac{-5}{6}$$

$$5\frac{5}{6} + \left(-\frac{9}{10}\right) = \frac{74}{15}$$

$$\frac{62}{15} + \frac{19}{10} = \frac{181}{30}$$

7.  $1\frac{2}{3} + \left(-\frac{5}{2}\right) = \frac{-5}{6}$

9.  $\left(-\frac{7}{4}\right) + \left(-\frac{23}{5}\right) = \frac{-127}{20}$

8.  $2\frac{1}{3} + \frac{13}{5} = \frac{74}{15}$

10.  $14\frac{5}{6} + \left(-\frac{44}{5}\right) = \frac{181}{30}$

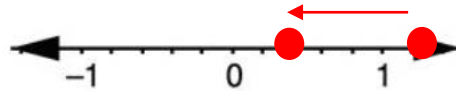
Grade 6 Accelerated  
Unit 11  
Lesson 2 Homework

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Find the difference using a number line representation for each phrase for questions 1-2.

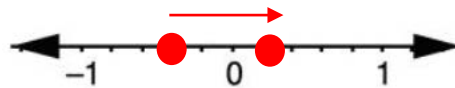
1. The difference of  $\frac{2}{5}$  and  $\frac{6}{5}$ .

$$\frac{2}{5} - \frac{6}{5} = -\frac{4}{5}$$



2. The difference of 0.2 and  $-0.4$ .

$$0.2 - (-0.4) = 0.6$$



Find the difference of each expression in questions 3-6.

3.  $14.75 - (-6.82)$

$$14.75 + 6.82 = 21.57$$

5.  $\left(-\frac{14}{8}\right) - \left(5\frac{5}{12}\right)$

$$-7\frac{1}{6} \text{ or } \frac{-43}{6}$$

4.  $\left(-5\frac{4}{8}\right) - \left(3\frac{2}{3}\right)$

$$-9\frac{1}{6} \text{ or } \frac{-55}{6}$$

6.  $-14.37 - (-8.54)$

$$-14.37 + 8.54 = -5.83$$

7. What is the difference of  $\frac{82}{12}$  and  $4\frac{4}{6}$ ?

$$\frac{82}{12} - \frac{28}{6} = \frac{13}{6} \text{ or } 2\frac{1}{6}$$

8. What is the difference of 5.89 and 13.19?

$$5.89 - 13.19 = -7.3$$

9. What is the difference of  $-26.1$  and  $-14.07$ ?

$$-26.1 - (-14.07) = -12.03$$

10. What is the difference of  $25\frac{3}{4}$  and  $\frac{21}{6}$ ?

$$\frac{103}{4} - \frac{21}{6} = \frac{89}{4} \text{ or } 22\frac{1}{4}$$

Grade 6 Accelerated  
Unit 11  
Lesson 3 Homework

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Identify if the product will be positive or negative for questions 1-3.

| Expression                                              | Sign of Product |
|---------------------------------------------------------|-----------------|
| 1. $\left(\frac{5}{8}\right)\left(-3\frac{1}{2}\right)$ | negative        |
| 2. $(-12.3)(-4.1)$                                      | positive        |
| 3. $\left(-\frac{5}{3}\right)\left(\frac{3}{6}\right)$  | negative        |

Find the product for questions 4-10.

4.  $(3.12)(-8.4) = -26.208$

5.  $\left(1\frac{5}{6}\right)\left(2\frac{3}{4}\right)$

$$\left(\frac{11}{6}\right)\left(\frac{11}{4}\right) = \frac{121}{24} = 5\frac{1}{24}$$

6.  $\left(-\frac{8}{3}\right)\left(\frac{9}{5}\right) = -\frac{72}{15} = -\frac{24}{5} = -4\frac{4}{5}$

$$7. \quad (-4.2)(-8.65) = 36.33$$

$$8. \quad (-0.15)(8.7) = -1.305$$

$$9. \quad \left(-\frac{4}{6}\right)\left(-\frac{14}{8}\right)$$

$$\frac{56}{48} = \frac{28}{24} = \frac{7}{6} = 1\frac{1}{6}$$

$$10. \quad (5.97)(3.1) = 18.507$$

Grade 6 Accelerated  
Unit 11  
Lesson 4 Homework

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Identify if the quotient will be positive or negative for questions 1-2.

| Expression                              | Sign of Quotient |
|-----------------------------------------|------------------|
| 1. $(-4.2) \div (3.15)$                 | Negative         |
| 2. $(-\frac{4}{5}) \div (-\frac{3}{2})$ | Positive         |

Find the quotient for questions 3-10.

3.  $(-7.632) \div (0.8) = -9.54$

$0.8 \overline{) 7.632} \longrightarrow$

$$\begin{array}{r} 9.54 \\ 8 \overline{) 76.32} \\ \underline{-72} \phantom{00} \\ 43 \phantom{00} \\ \underline{-40} \phantom{00} \\ 32 \phantom{00} \\ \underline{-32} \phantom{00} \\ 0 \end{array}$$

4.  $(\frac{49}{15}) \div (-\frac{7}{10})$

$$\frac{49}{15} \cdot (-\frac{10}{7}) = \frac{-490}{105} = -\frac{14}{3} = -4\frac{2}{3}$$

5.  $(-\frac{33}{7}) \div (-5\frac{3}{6})$

$$-\frac{33}{7} \cdot (-\frac{6}{33}) = \frac{6}{7}$$

6.  $(24.48) \div (3.2) = 7.65$

$3.2 \overline{) 24.48} \longrightarrow$

$$\begin{array}{r} 7.65 \\ 32 \overline{) 244.80} \\ \underline{-224} \phantom{00} \\ 208 \phantom{00} \\ \underline{-192} \phantom{00} \\ 160 \phantom{00} \\ \underline{-160} \phantom{00} \\ 0 \end{array}$$

$$7. \quad \left(-\frac{27}{5}\right) \div \left(3\frac{3}{5}\right)$$

$$\frac{-27}{5} \div \frac{18}{5} = \frac{-27}{5} \cdot \frac{5}{18} = -\frac{27}{18} = -\frac{9}{6} = -\frac{3}{2} = -1\frac{1}{2}$$

$$8. \quad (-46.28) \div (-5.2) = 8.9$$

$$5.2 \overline{) 46.28} \longrightarrow \begin{array}{r} 8.9 \\ 52 \overline{) 462.8} \\ \underline{-416} \phantom{0} \\ 468 \\ \underline{-468} \\ 0 \end{array}$$

$$9. \quad \left(10\frac{2}{3}\right) \div \left(-\frac{5}{2}\right)$$

$$\frac{32}{3} \cdot \left(-\frac{2}{5}\right) = \frac{-64}{15} = -4\frac{4}{15}$$

$$10. \quad (5.84) \div (0.8) = 7.3$$

$$0.8 \overline{) 5.84} \longrightarrow \begin{array}{r} 7.3 \\ 8 \overline{) 58.4} \\ \underline{-56} \phantom{0} \\ 24 \\ \underline{-24} \\ 0 \end{array}$$



Use the table below to answer questions 1-3

The table shows transactions for 3 different checking accounts. A negative number represents a debit and a positive number represents a credit.

| Account 1 | Account 2 | Account 3 |
|-----------|-----------|-----------|
| \$225.90  | \$318.25  | \$253.30  |
| -\$31.65  | \$22.15   | -\$112.25 |
| \$98.70   | -\$102.40 | \$89.10   |
| \$45.90   | -\$19.70  | -\$14.15  |

1. What is the ending balance of Account 1?

$$225.90 - 31.65 + 98.70 + 45.90 = \$338.85$$

2. What is the ending balance of Account 2?

$$318.25 + 22.15 - 102.40 - 19.70 = \$218.30$$

3. What is the ending balance of Account 3?

$$253.30 - 112.25 + 89.10 - 14.15 = \$216.00$$

Use the information below to answer questions 4-5.

Foster recently went on a weekend hiking trip and recorded the elevation changes at several checkpoints. Foster's elevation log can be seen in the table below.

| Checkpoint | Beginning Elevation (ft) | Final Elevation (ft) |
|------------|--------------------------|----------------------|
| 1          | -125.4                   | -78                  |
| 2          | -78                      | -101.6               |
| 3          | -101.6                   | 14.7                 |
| 4          | 14.7                     | 225.1                |

Final – Beginning

47.4  
-23.6  
116.3  
210.4

4. Which checkpoint at the greatest change in elevation?

4

5. If the next checkpoint has a change a final elevation of 148.3, what is the change in elevation?

$$148.3 - 225.1 = -76.8 \text{ ft.}$$

6. Stanley's office has a parking garage that charges \$12.75 each day for parking. Stanley's employer provides him with a parking card that has \$250 each month. When Stanley parked today, the card indicated it had \$84.25 left on the card. How many days has Stanley used the card for parking this month?

$$\frac{250 - 84.25}{12.75} = 13 \text{ days}$$

7. Tanner is traveling on his first flight to visit family.  $9\frac{3}{5}$  minutes into the flight, the pilot announces the plane is currently at an altitude of 6.15 miles. What was the rate, in minutes, the plane reached to this cruising altitude?

$$\frac{6.15 \text{ miles}}{9\frac{3}{5} \text{ min}} = \frac{x \text{ miles}}{1 \text{ min}} \quad x \approx 0.64 \text{ miles} \quad \text{Rate} = 0.64 \text{ miles per minute}$$

8. A penguin sitting on an iceberg notices a fish swimming below. The fish is 15.6 ft below sea level. The penguin dives and reaches the fish in  $\frac{5}{8}$  of a minute. What was the rate of descent the penguin made?

$$\frac{15.6 \text{ ft}}{\frac{5}{8} \text{ min}} = \frac{x \text{ ft}}{1 \text{ min}} \quad x = 24.96 \quad \text{Rate} = 24.96 \text{ ft per minute}$$

9. The temperature in Alaska today was  $-58.8^\circ\text{C}$ . Over the next 4 weeks, the temperature will increase an average of  $3.6^\circ\text{C}$  each week. What will the temperature be at the end of the 4 weeks?

$$-58.8 + (4 \cdot 3.6) = -44.4^\circ\text{C}$$

10. Phoebe buys her dogs a bag of food that is 80.6 ounces. Each of Phoebe's dogs eat  $\frac{7}{16}$  ounces a day. If she has 3 dogs, how long will the bag of dog food last Phoebe?

$$\frac{80.6}{3 \cdot \frac{7}{16}} \approx 61 \text{ days}$$

Grade 6 Accelerated  
Unit 11  
Lesson 6 Homework

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

1. Sophia purchased a new car for \$26,714.00. She read that the value of new cars change by  $\frac{-4}{19}$  in the first year. What is the value of Sophia's car after the first year?

$$26,714 \left( \frac{15}{19} \right) = \$21,090 \text{ is the value after the first year.}$$

2. Sophia learns that the value of her car will continue to change by  $\frac{-3}{17}$  of its value each year after the first year. What is the value of Sophia's car at the end of year 2?

$$21,090 \left( \frac{14}{17} \right) = \$17,368.24 \text{ is the value at the end of the second year.}$$

3. An Orca whale is swimming  $-250\frac{3}{4}$  ft below sea level when he notices the rest of his pod above him at the surface. Orca whales can swim at a rate of 183.74 feet per hour. How long, in hours, will it take the Orca to reach his pod?

$$\frac{183.74 \text{ ft}}{1 \text{ hr}} = \frac{-250\frac{3}{4} \text{ ft}}{x \text{ hr}} \quad x \approx 1.4 \text{ hours}$$

4. A candle store sells the 3-wick candles for  $3\frac{1}{3}$  times what they paid for them from the warehouse. The candles are purchased from the warehouse at \$6.75. How much profit does the candle store make off each candle?

$$6.75 \times \frac{10}{3} = \$22.50 \quad 22.50 - 6.75 = \$15.75 \text{ profit}$$

5. After 6 months, the price of the candles is reduced by  $\frac{1}{5}$  of the original store price. What is the price of the candles after the 6 month reduction?

$$22.50 - \left( \frac{1}{5} \cdot 22.50 \right) = \$18.00$$

6. Priscilla is planning to hike a new trail this weekend. She sets up camp Friday evening at the start of the trail which begins at an elevation of  $-184.56$  ft and ends at an elevation of  $759.22$  ft. She plans to begin her hike in the early morning and wants to travel  $\frac{5}{8}$  of the change in elevation of her trip on Saturday, then set up camp until the next morning. What is Priscilla's starting elevation on Sunday?

$$759.22 - (-184.56) = 943.78$$

$$\frac{5}{8}(943.78) = 589.86 \text{ ft. on Sat.}$$

$$-184.56 + 589.86 = 405.3 \text{ ft is the starting elevation on Sunday.}$$

7. Kate is running a race on Monday, which has a total distance of  $26.13$  miles. Kate's family wants to show their support by holding up encouraging signs along the path. Her mother decides to stand at the spot that is  $\frac{4}{13}$  of the total race while her father chooses to stand at the spot that is  $\frac{10}{13}$  of the total race. How many miles apart are Kate's parents standing from each other?

$$26.13 \left( \frac{4}{13} \right) = 8.04 \text{ miles}$$

$$20.1 - 8.04 = 12.06 \text{ miles apart}$$

$$26.13 \left( \frac{10}{13} \right) = 20.1 \text{ miles}$$

8. Kate runs at a steady rate of  $12.3$  minutes per mile. Using the information given in Question 7, how many minutes after the race starts should her mother expect to see Kate run by?

$$\frac{8.04 \text{ miles}}{x \text{ min}} = \frac{1 \text{ mile}}{12.3 \text{ min}} \quad x = 12.3(8.04) = 98.892 \text{ minutes}$$

9. Joshua used his store credit card to purchase a new basketball. He has a coupon that when he uses his credit card, the item will be  $\frac{-1}{3}$  of the total. The basketball was priced at  $\$21.39$ . A month goes by and Joshua forgot to pay his credit card bill, so they have added a fee of  $0.29$  times the amount he owed. What is the total amount Joshua has to pay his credit card company?

$$21.39 \left( \frac{2}{3} \right) = 14.26$$

$$\text{Joshua owes } \$18.40$$

$$14.26 \cdot 1.29 = 18.40$$

10. Sandy is moving to Jacksonville, Florida from New York, New York. The total distance of her move is  $945.61$  miles. She rented a moving truck that charges  $\$19.95$  a day and  $\$0.35$  per mile over  $500$  miles. If she returns the truck with a full tank of gas, the company will discount her bill by  $\frac{-2}{31}$ . What is the total amount Sandy will pay for the moving truck if she fills the tank with gas before returning it?

Assuming she drives it all in one day:

$$19.95 + 0.35(945.61 - 500) = 175.91$$

$$175.91 \left( \frac{29}{31} \right) = \$164.56 \text{ is Sandy's total bill for the moving truck.}$$

Grade 6 Accelerated  
Unit 11  
Lesson 7 Homework

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Use the scenario for questions 1-3.

Pierson and his friends plan to use a rideshare service for his birthday excursion to International Drive in Orlando, Florida. The service has a booking fee of \$5.75 and charges \$0.46 for the first  $\frac{1}{2}$  mile and \$0.26 for each additional 0.2 mile after the first  $\frac{1}{2}$  mile.

1. How much will Pierson pay if his trip to International Drive is 23.45 miles from his home?

$$\begin{array}{r} 23.45 \\ -0.5 \\ \hline 22.95 \end{array} \quad \begin{array}{l} \$5.75 + \$0.46 + \$0.26 \left( \frac{22.95}{0.2} \right) \\ \$5.75 + \$0.46 + \$0.26(114.75) \end{array} \quad \begin{array}{l} \$5.75 + \$0.46 + \$29.835 \\ = \$36.05 \text{ total} \end{array}$$

2. Pierson decides to pick up an additional friend. When they arrive at the friend's house, she is not ready to leave. The rideshare service charges \$0.75 for each minute they have to wait. The total bill was \$44.30. How long did Pierson's friend take to get ready?

$$\begin{array}{r} 44.30 \\ -36.05 \\ \hline 8.25 \end{array} \quad \frac{8.25}{0.75} = 11 \quad \text{The friend took 11 minutes to get ready.}$$

3. Pierson's friends decide to use the same rideshare service to take everyone home. They drop the first friend off at his home 19.6 miles from International Drive. The second friend lives another 2.42 miles and Pierson lives another 6.8 miles away. He decides to tip the driver  $\frac{2}{9}$  of the bill. How much did Pierson pay for the service, including tip?

$$\begin{array}{r} 19.6 \\ +2.42 \\ \hline 22.02 \\ +6.8 \\ \hline 28.82 \end{array} \quad \begin{array}{l} 28.82 \\ -0.50 \\ \hline 28.32 \end{array} \quad \begin{array}{l} 5.75 + 0.46 + 0.26 \left( \frac{28.32}{0.2} \right) \\ 5.75 + 0.46 + 36.816 \\ 43.03 \text{ plus tip} \Rightarrow 43.03 \cdot 1\frac{2}{9} = \$52.29 \text{ total with tip} \end{array}$$

4. Ian purchased  $1\frac{5}{6}$  pounds of tomatoes,  $3\frac{1}{3}$  pounds of onions, 5.3 pounds of avocado and 2 limes to make homemade guacamole.

| Produce  | Price (per pound) |
|----------|-------------------|
| Tomatoes | \$0.89 per pound  |
| Onions   | \$0.65 per pound  |
| Lime     | \$0.75 each       |
| Avocado  | \$1.09 per pound  |

Based on the listed prices, how much would Ian spend?

$$1\frac{5}{6} \cdot 0.89 = 1.63 \quad 3\frac{1}{3} \cdot 0.65 = 2.17 \quad 5.3 \cdot 1.09 = 5.78 \quad 2 \cdot 0.75 = 1.50$$

Ian spent \$11.08 total.

Use the scenario for questions 5-7.

Ethan offers a tutoring service for elementary students. He charges an initial fee of \$12.50 for the first tutoring session and provides service for 1.5 hours. After the first session, he charges \$6.75 for  $\frac{3}{4}$  hour and charges \$1.86 for each additional  $\frac{1}{6}$  hour after the first  $\frac{3}{4}$  hour.

5. Ethan tutored a new client this week. The first session was held on Monday, then tutored the client on Wednesday and received payment totaling \$38.78. How many hours did he tutor on Wednesday?

$$\begin{array}{r} 38.78 \\ -12.50 \\ \hline 26.28 \end{array} \quad \begin{array}{r} 26.28 \\ -6.75 \\ \hline 19.53 \end{array} \left( \frac{3}{4} \text{ hour} \right) \quad \frac{19.53}{1.86} = 10.5 \frac{1}{6} \text{ hour} \quad \frac{3}{4} + \left( 10 \frac{1}{2} \cdot \frac{1}{6} \right) = 2.5 \text{ hours}$$

6. Ethan tutored a second new client this same week. The first session was held on Tuesday, then tutored the client on Thursday for 1.6 hours. On Thursday, how much would he receive for the total tutoring of this client?

$$\begin{array}{r} 1.6 \\ -0.75 \\ \hline 0.85 \text{ hours} \end{array} \quad \frac{0.85}{\frac{1}{6}} = 5.1 \quad \begin{array}{r} 12.50 + 6.75 + 5.1(1.86) \\ 12.50 + 6.75 + 9.486 \\ \hline \$28.74 \end{array}$$

7. If Ethan has a third new client this week and only provides the initial tutoring session, how much money in total has he made for the week for all three clients?

$$38.78 + 28.74 + 12.50 = \$80.02 \quad \text{Ethan made a total of } \$80.02$$

8. A candy store keeps track of the weekly changes in their inventory. Candy is delivered to the shop in  $25\frac{1}{8}$ -pound (lb) bags, 16 bags at a time twice a week. The inventory at the end of week shows a total of  $133\frac{3}{4}$  lbs. If the same amount of candy was sold each day, how many pounds of candy did they sell in one day?

$$32 \cdot 25\frac{1}{8} = 804 \text{ total bags (lbs. received)} \quad \frac{670.25}{7} = 95.75 \text{ lbs. were sold each day}$$

$$804 - 133\frac{3}{4} = 670.25 \text{ lbs. sold}$$

9. Charlie is building birdhouses for the neighborhood cardinals. To make one birdhouse, she needs 4 pieces of wood  $\frac{5}{6}$  feet for the walls, 1 piece of wood 0.33 feet for the floor and 2 pieces of wood  $\frac{5}{12}$  feet for the roof. A single board is 9 feet long and costs \$9.32. If Charlie has \$30, how many birdhouses can she make?

$$4 \left( \frac{5}{6} \right) + 0.33 + 2 \left( \frac{5}{12} \right) = \frac{10}{3} + 0.33 + \frac{5}{6} = \frac{1349}{300} \text{ ft./house} \quad \frac{30}{9.32} = 3 \text{ boards she can buy}$$

$$9 \div \frac{1349}{300} = \frac{2700}{1349} = 2 \frac{2}{1349} \text{ birdhouses per board} \quad 3 \cdot 2 \frac{2}{1349} \approx 6 \text{ birdhouses}$$

10. Camila is making bows to sell to support a local nonprofit organization. She purchased a roll of ribbon that is 38.75 ft in length and each bow uses  $2\frac{7}{12}$  feet to make. If she makes 11 bows, how many more bows can be made with the ribbon left in the roll?

$$11 \cdot 2\frac{7}{12} = 28\frac{5}{12} \text{ ft used} \quad 10\frac{1}{3} \div 2\frac{7}{12} = 4$$

$$38.75 - 28\frac{5}{12} = 10\frac{1}{3} \text{ ft left} \quad \text{Camila can make 4 more bows.}$$

Grade 6 Accelerated  
Unit 11  
Lesson 8 Homework

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

1. Emily says that when she evaluated an expression, the result was 51. Use mental math to estimate each value. Then, circle which of the following expressions could have been Emily's expression.

$$(8 + 3)^2 - 22$$

$$8^2 + 3^2 - 22$$

$$(8 - (-3)^2) + 22$$

$$11^2 + 22$$

2. Cindy evaluated the expression  $5^2 \div 5(3.25 + 5.25)$ .

Circle Cindy's error and choose which step of the order of operations she did not apply correctly.

| Cindy's Steps                                                     | Order of Operations                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $5^2 \div 5(3.25 + 5.25)$<br>$10 \div 5(8.5)$<br>$2(8.5)$<br>$17$ | <ul style="list-style-type: none"> <li>Evaluate expressions with exponents</li> <li>Perform operations inside grouping symbols.</li> <li>Perform multiplication and division in order from left to right.</li> <li>Perform addition and subtraction in order from left to right.</li> </ul> |

3. Evaluate the expression without using a calculator. Write an explanation using the order of operations next to each step.

| $ 4 - 12  \div (-8)$ | Explanation     |
|----------------------|-----------------|
| $ -8  \div (-8)$     | Grouping symbol |
| $8 \div (-8)$        | Division        |
| $-1$                 |                 |

4. Evaluate the expression without using a calculator. Write an explanation using the order of operations next to each step.

| $(-3.5 \times 4)^2 \div 7$ | Explanation     |
|----------------------------|-----------------|
| $(-14)^2 \div 7$           | Grouping symbol |
| $196 \div 7$               | Exponents       |
| 28                         | Division        |

Evaluate each expression.

5.  $\frac{(-14 \times 0.5)^3}{14} = \frac{-7^3}{14} = \frac{-343}{14} = -24.5$

6.  $6 \times |-3.4 \times 5.2| \quad 6 \times |-17.68| = 6 \times 17.68 = 106.08$

7.  $(-84 \times 0.6)^2 + 6 \quad (-50.4)^2 + 6 = 2540.16 + 6 = 2546.16$

8.  $[(16 - 5 \times 4) \times 3]^3$

$$[(16 - 20) \times 3]^3$$

$$[-4 \times 3]^3$$

$$-12^3$$

$$-1,728$$

9.  $64 \div [(-3)^2 \times 4 - 2]$

$$64 \div [9 \times 4 - 2]$$

$$64 \div [36 - 2]$$

$$64 \div 34$$

$$1 \frac{15}{17}$$

10.  $|-6.3 \times 5| - 5^2$

$$|-31.5| - 25$$

$$31.5 - 25$$

$$6.5$$



Grade 6 Accelerated  
Unit 11  
Lesson 9 Homework

Name \_\_\_\_\_ Answer Key \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

In the table, an expression has been evaluated, but the steps are mixed up. Order the expression using the numbers 1-5 on the left, the final answer has been given to you. Write an explanation for each step.

| Step   | Expression                        | Explain the Step               |
|--------|-----------------------------------|--------------------------------|
| 4      | $[5]^3 \div 25$                   | Addition within grouping       |
| 2      | $[16 \times 0.25 + 1]^3 \div 25$  | Exponents within grouping      |
| 3      | $[4 + 1]^3 \div 25$               | Multiplication within grouping |
| 1      | $[4^2 \times 0.25 + 1]^3 \div 25$ | Original expression            |
| 5      | $125 \div 25$                     | Exponents                      |
| Answer | 5                                 | Divison                        |

Evaluate each expression without using a calculator. Show work to support your answer.

6.  $\left(\frac{1}{2}\right)^2 \times 36 + 6^2$

$$\begin{aligned} &\frac{1}{4} \times 36 + 36 \\ &9 + 36 \\ &45 \end{aligned}$$

7.  $[(16 + 1.5)^2 - 6.25]^2 \div 100$

$$\begin{aligned} &[(17.5)^2 - 6.25]^2 \div 100 \\ &[306.25 - 6.25]^2 \div 100 \\ &[300]^2 \div 100 \\ &90,000 \div 100 \\ &900 \end{aligned}$$

8.  $250 - (2^3 - 2)^3 \times 2$

$$\begin{aligned} &250 - (8 - 2)^3 \times 2 \\ &250 - (6)^3 \times 2 \\ &250 - 216 \times 2 \\ &250 - 432 \\ &-182 \end{aligned}$$

9.  $-|-9 + \frac{1}{2}| + \frac{2}{3}(8^2 - 7^1)$

$$\begin{aligned} &-|-8.5| + \frac{2}{3}(64 - 7) \\ &-8.5 + \frac{2}{3}(57) \\ &-8.5 + 38 \\ &29.5 \end{aligned}$$

10.  $\frac{(-12.5 + |-9.25 - 4.25|)^4 + 31}{16}$

$$\begin{aligned} &\frac{(-12.5 + |-13.5|)^4 + 31}{16} \\ &\frac{(-12.5 + 13.5)^4 + 31}{16} = \frac{1^4 + 31}{16} = \frac{1 + 31}{16} = \frac{32}{16} \\ &2 \end{aligned}$$